Attention Is All You Need: Scalable Sequence Trans duction Models

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ABSTRACT

The dominant sequence transduction models are based on complex recurrent or convolutional neural networks in an encoder-decoder configuration. In this work, we propose the Transformer, a model architecture eschewing recurrence and instead relying entirely on an attention mechanism to draw global dependencies between input and output. The Transformer allows for significantly more parallelization and can reach a new state of the art in translation quality after being trained for as little as twelve hours on eight P100 GPUs. Experiments on two machine translation tasks show these models to be superior in quality while being more parallelizable and requiring significantly less time to train.

1. INTRODUCTION

Recurrent neural networks, especially long short-term memory (LSTM) and gated recurrent units (GRU), have been firmly established as state-of-the-art approaches in sequence modeling and transduction problems such as language modeling and machine translation. Numerous efforts have since continued to push the boundaries of recurrent language models and encoder-decoder architectures. Recurrent models typically factor computation along the symbol positions of the input and output sequences. Aligning the positions to steps in computation time, they generate a sequence of hidden states h_t, as a function of the previous hidden state h_{t-1} and the input for position t. This inherently sequential nature precludes parallelization within training examples, which becomes critical at longer sequence lengths, as memory constraints limit batching across examples.

2. PROPOSED METHODOLOGY & SYSTEM ARCHITECTURE

The Transformer follows this overall architecture using stacked self-attention and point-wise, fully connected layers for both the encoder and decoder. The encoder is composed of a stack of N = 6 identical layers. Each layer has two sub-layers: a multi-head self-attention mechanism, and a simple, position-wise fully connected feed-forward network. We employ a residual connection around each of the two sub-layers, followed by layer normalization. To compute attention, we map a query and a set of key-value pairs to an output. Scaled Dot-Product Attention computes the dot products of the query with all keys, divides each by the square root of the key dimension d_k, and applies a softmax function to obtain weights on the values.

3. EXPERIMENTAL EVALUATION & RESULTS

On the WMT 2014 English-to-German translation task, the big transformer model achieves a state-of-the-art BLEU score of 28.4, improving over the existing best results, including ensembles by more than 2.0 BLEU. On the WMT 2014 English-to-French translation task, our big model achieves a BLEU score of 41.0, outperforming all of the previously published single models, at 1/4 the training cost of the previous state-of-the-art model. Model training was exceptionally fast, establishing new records on both benchmark suites.

4. CONCLUSION AND FUTURE WORK

In this work, we presented the Transformer, the first sequence transduction model based entirely on attention, replacing the recurrent layers most commonly used in encoder-decoder architectures with multi-headed self-attention. For translation tasks, the Transformer can be trained significantly faster than architectures based on recurrent or convolutional layers. On both WMT 2014 English-to-German and WMT 2014 English-to-French translation tasks, we achieve a new state of the art.